

SPECIALIST READING

A

Find the answers to these questions in the following texts.

- 1 What is one of the main causes of a PC not running at its highest potential speed?
- 2 What word in the text is used instead of 'buffer'?
- 3 What device looks after cache coherency?
- 4 What is the main alternative to 'write-through cache'?
- 5 When does a write-back cache write its contents back to main memory?
- 6 When is data marked as 'dirty' in a write-back cache?
- 7 What determines what data is replaced in a disk cache?

CACHE MEMORY

Most PCs are held back not by the speed of their main processor, but by the time it takes to move data in and out of memory. One of the most important techniques for getting around this

bottleneck is the memory cache.

The idea is to use a small number of very fast memory chips as a buffer or cache between main memory and the processor. Whenever the processor needs to read data it looks in this cache area first. If it finds the data in the cache then this counts as a 'cache hit' and the processor need not go through the more laborious process of reading data from the main memory. Only if the data is not in the cache does it need to access main

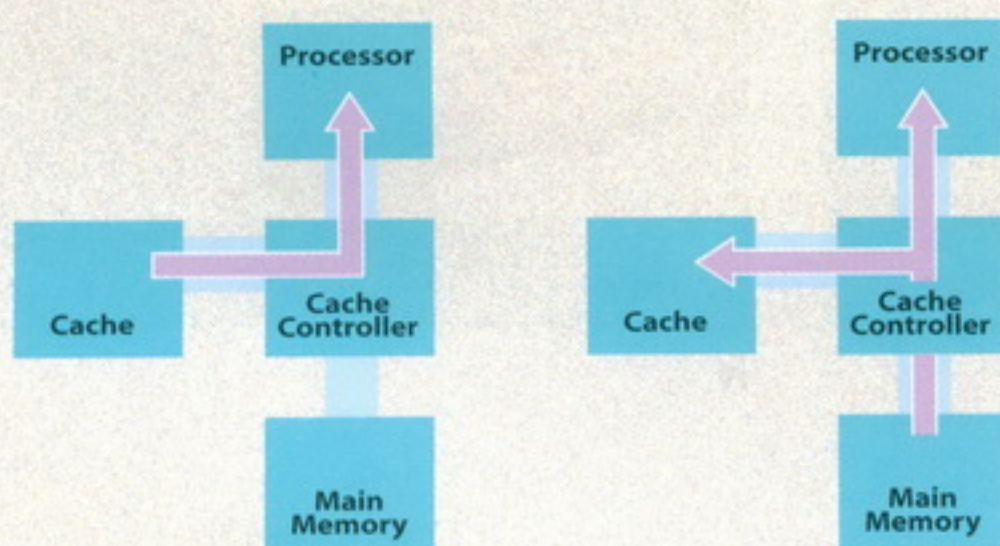
memory, but in the process it copies whatever it finds into the cache so that it is there ready for the next time it is needed. The whole process is controlled by a group of logic circuits called the cache controller.

One of the cache controller's main jobs is to look after 'cache coherency' which means ensuring that any changes written to main memory are reflected within the cache and vice versa. There are several techniques for achieving this, the most obvious

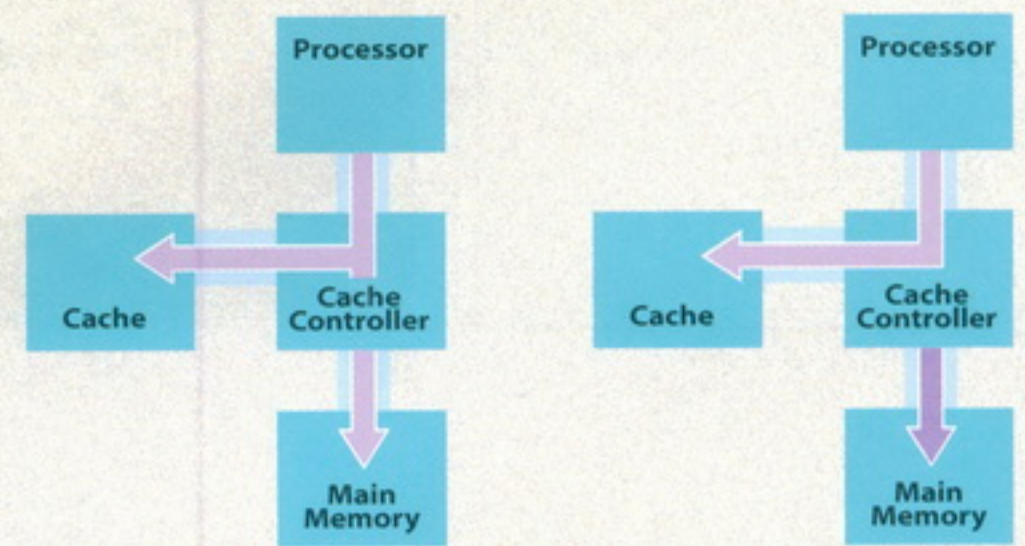
being for the processor to write directly to both the cache and main memory at the same time. This is known as a 'write-through' cache and is the safest solution, but also the slowest.

The main alternative is the 'write-back' cache which allows the processor to write changes only to the cache and not to main memory. Cache entries that have changed are flagged as 'dirty', telling the cache controller to write their contents back to main memory before using the space to cache new data. A write-back cache speeds up the write process, but does require a more intelligent cache controller.

Most cache controllers move a 'line' of data rather than just a single item each time they need to transfer data between main memory and the cache. This tends to improve the chance of a cache hit as most programs spend their time stepping through instructions stored sequentially in memory, rather than jumping about from one area to another. The amount of data transferred each time is known as the 'line size'.



If there is a cache hit then the processor only needs to access the cache. If there is a miss then it needs to both fetch data from main memory and update the cache, which takes longer. With a standard write-through cache, data has to be written

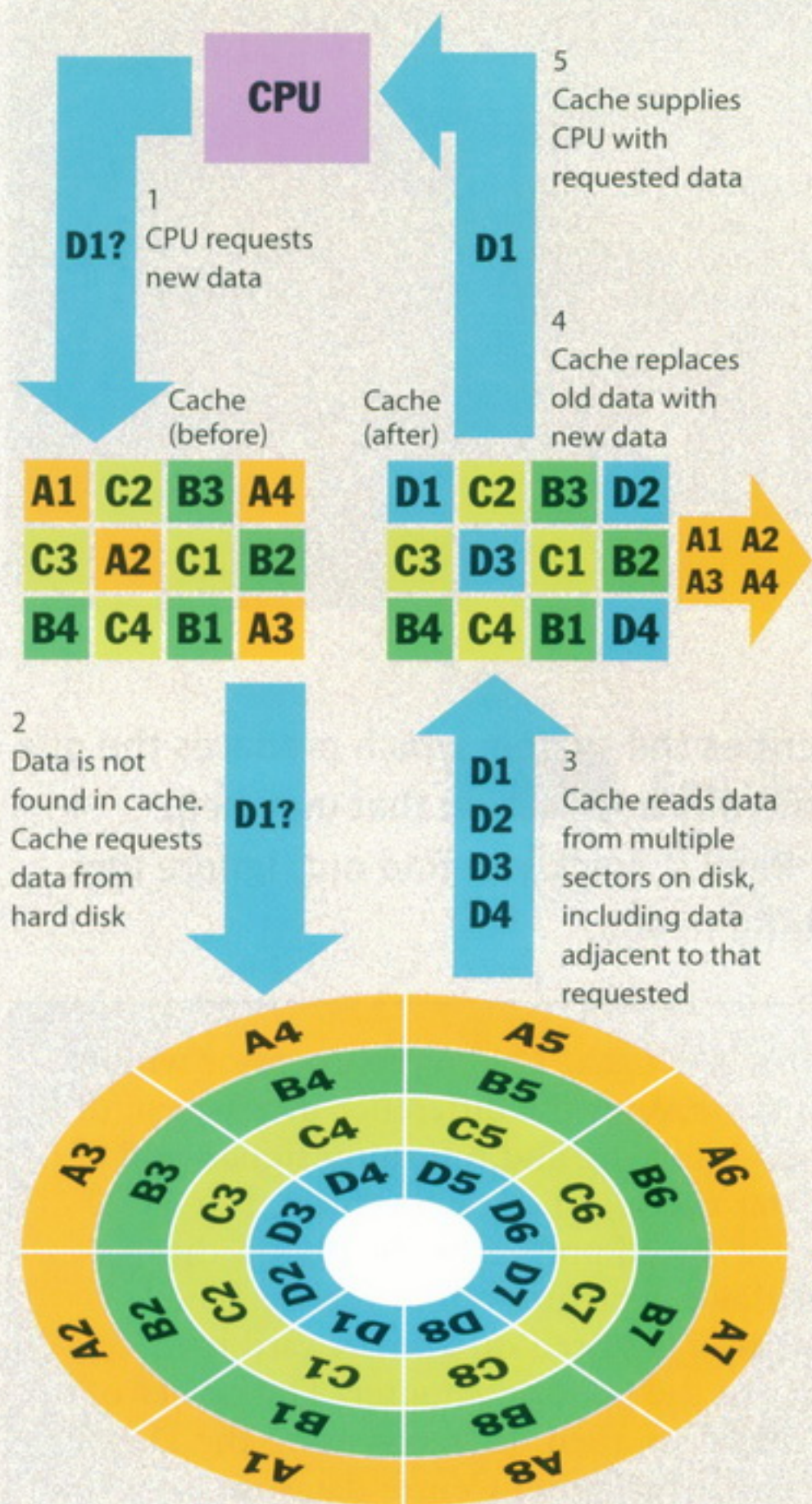


both to main memory and to the cache. With a write-back cache the processor needs only write to the cache, leaving the cache controller to write data back to main memory later on.

How a Disk Cache Works

Disk caching works in essentially the same way whether you have a cache on your disk controller or you are using a software-based solution. The CPU requests specific data from the cache. In some cases, the information will already be there and the request can be met without accessing the hard disk.

If the requested information isn't in the cache, the data is read from the disk along with a large chunk of adjacent information. The cache then makes room for the new data by replacing old. Depending on the algorithm that is being applied, this may be the information that has been in the cache the longest, or the information that is the least recently used. The CPU's request can then be met, and the cache already has the adjacent data loaded in anticipation of that information being requested next.



B Re-read the texts to find the answers to these questions.

1 Match the terms in Table A with the statements in Table B.

Table A

- a Cache hit
- b Cache controller
- c Cache coherency
- d Write-through cache
- e Write-back cache
- f Line size

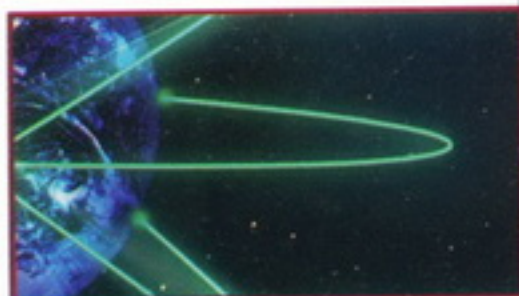
Table B

- i The process of writing changes only to the cache and not to main memory unless the space is used to cache new data
- ii The amount of data transferred to the cache at any one time
- iii The process of writing directly to both the cache and main memory at the same time
- iv The processor is successful in finding the data in the cache
- v Ensuring that any changes written to main memory are reflected within the cache and vice versa
- vi The logic circuits used to control the cache process

2 Mark the following as True or False:

- a Cache memory is faster than RAM.
- b The processor looks for data in the main memory first.
- c Write-through cache is faster than write-back cache.
- d Write-back cache requires a more intelligent cache controller.
- e Most programs use instructions that are stored in sequence in memory.
- f Most cache controllers transfer one item of data at a time.
- g Hardware and software disk caches work in much the same way.

[Adapted from 'How a Disk Cache Works', PC Magazine, September 1990]



Computer Applications

STARTER

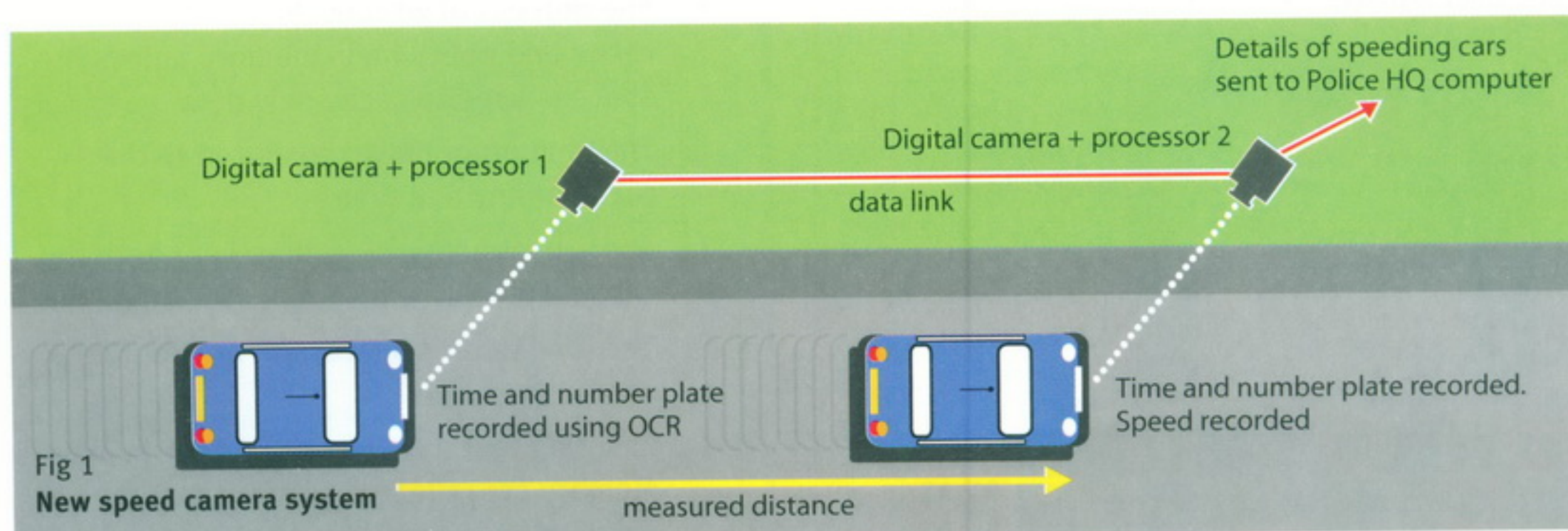
1 Work in groups. List as many uses as you can for computers in one of these areas.

- 1 supermarkets
- 2 hospitals
- 3 airports
- 4 police headquarters

READING

2 Study this diagram. Using only the diagram, try to list each stage in the operation of this computerised speed trap to make an explanation of how it operates. For example:

- 1 Camera 1 records the time each vehicle passes.



3 Part 1 of the text describes the system which predates the one shown in Fig 1. Does it contain any information that may help complete your explanation? Read it quickly to find out. Ignore any information which is not helpful to you.

Part 1

In the last ten years, police have installed speed trap units on many busy roads. These contain a radar set, a microprocessor and a camera equipped with a flash. The radar sends out a beam of radio waves at a frequency of 24 gigahertz. This is equivalent to a wavelength of 1.25 cms. If a car is moving towards the radar, the reflected signal will bounce back with a slightly smaller wavelength. If away from the radar, the waves will reflect with a slightly

longer wavelength. The microprocessor within the unit measures the difference in wavelength between outgoing and returning signals and calculates the speed of each vehicle. If it is above the speed pre-set by the police, the camera takes a picture of the vehicle. The information is stored on a smart card for transfer to the police computer. The owner of the vehicle can then be traced using the Driver and Vehicle Licensing Centre database.

4 Part 2 describes the new system. Read it to complete the stages in your explanation.

Part 2

Some drivers have now got used to these traps. They slow down when they approach one to ensure that the camera is not triggered. They speed up again as soon as they have passed. This is known as 'surfing'. One way of outwitting such motorists is a new computerised system. This consists of two units equipped with digital cameras positioned at a measured distance apart. The first unit records the time each vehicle passes it and identifies each vehicle by its number plates

using optical character recognition software. This information is relayed to the second unit which repeats the exercise. The microprocessor within the second unit then calculates the time taken by each vehicle to travel between the units. The registration numbers of those vehicles exceeding the speed limit are relayed to police headquarters where a computer matches each vehicle with the DVLC database. Using mailmerge a standard letter is then printed off addressed to the vehicle owner.

LANGUAGE WORK

Present passive

Study these sentences.

- 1 The radar sends out a beam of radio waves.
- 2 The information is stored on a smart card.

In 1 the verb is active and in 2 it is passive, the Present passive. Why is this so? What difference does it make? In 1 the agent responsible for the action is included – the radar. In 2 the agent is not included although

we know what it is – the microprocessor. The passive is often used to describe the steps in a process where the action is more important than the agent and where the agent is already known to the reader. If we need to add the agent, we can do so like this:

- 3 The information is stored on a smart card *by the microprocessor.*

5 Describe the operation of the new speed trap by converting each of these statements to the Present passive. Add information on the agent where you think it is necessary.

- 1 The first unit records the time each vehicle passes.
- 2 It identifies each vehicle by its number plates using OCR software.
- 3 It relays the information to the second unit.
- 4 The second unit also records the time each vehicle passes.
- 5 The microprocessor calculates the time taken to travel between the units.
- 6 It relays the registration numbers of speeding vehicles to police headquarters.
- 7 A computer matches each vehicle with the DVLC database.
- 8 It prints off a letter to the vehicle owners using mailmerge.